

Cost Effectiveness of Intensity Modulated Radiotherapy (IMRT) versus Conventional Radiotherapy for the Treatment of Head and Neck Cancers in India: A Study Protocol

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ABSTRACT

Background: The field of cancer cure is flooded with various newer technologies ranging from advanced form of surgeries to new drugs to novel forms of radiotherapy techniques. While the newer forms of radiotherapy i.e., Intensity modulated radiotherapy (IMRT) or 3-dimensional conformal radiotherapy (3D-CRT) may have the potential of providing improved health outcomes, but they require an additional cost. Thus, it becomes necessary to compare the costs and benefits of a new technology, before considering its inclusion in the current health system. The present study will be done to assess the cost effectiveness of IMRT as compared to the conventional 2 dimensional radiotherapy (2DRT) for the treatment of head and neck cancers (HNCs) in India.

Methods: The present study will be a model based effectiveness evaluation. The study will include collecting data on health system cost and out of pocket expenditure incurred on the treatment of HNC, followed by assessment of health outcomes in the form of survival rates and quality of life subsequent treatment with IMRT and 2DRT and finally the development of Markov Model for estimating the incremental cost effectiveness ratio. The present study will be undertaken in the Departments of Radiotherapy and Otolaryngology of a large tertiary care hospital in North India. Health system costs will be assessed using bottom up methodology. Cost of illness approach would be followed for interviewing patients for assessing out of pocket (OOP) expenditure. Patients will be recruited from the Radiotherapy Department, and will be followed up till one year for assessing quality of life weights. Systematic review cum meta-analysis of the existing randomised controlled trials comparing IMRT and 2DRT would be undertaken for assessing pooled survival rates. A Markov model based on the natural history of the progression of HNC will be developed, following societal perspective and lifetime study horizon.

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INTRODUCTION

Cancers of the head and neck region (HNC) are the most prevalent cancers in India contributing to 15% of all cancers.(1) Further, India is a home to 28% the global deaths and 71% of the mortality in the South East Asia region, due to cancer of the head and neck region. (1) As per GLOBOCAN estimates, India had 1.45 lakhs incident cases and 1.05 deaths due to HNC in 2012.(1) By 2035, burden in terms of both incidence and mortality due to HNC is expected to rise by 77% and 78% respectively.(1)

This increased burden along with an increase in demand for cancer care has become a challenge to the health system of India. Government of India is spending a huge amount of money both at the national and state level for providing cancer treatment. The budget allocation towards cancer-specific spending in India, has increased from INR 115 million in the sixth plan (1980- 85) to INR 28719 million and INR 60000 million in the XIth and XIIth five year plan respectively. (2, 3) Further, through the roll out of publicly sponsored health insurance schemes such as Rashtriya Swasthya Bima Yojana (RSBY) at the national level and other state level health insurance schemes of Maharashtra, Andhra Pradesh, Tamil Nadu, etc., a huge amount of exchequer's money has gone into providing cancer cure in secondary and tertiary care hospitals.(4-6) Also, state level initiatives such as 'Mukhya Mantri Punjab Cancer Raahat Kosh Scheme' from the state of Punjab provide cashless treatment of cancer in public and private sector hospitals.(7)

The field of cancer cure is flooded with various newer technologies ranging advanced form of surgeries to new drugs in chemotherapy regimens to newer forms of radiotherapy techniques. Radiotherapy is one of the technologies which have been in use for the last 30-40 years for the treatment of cancers. The recent innovations in this field, i.e., 3-dimensional conformal

radiotherapy (3D-CRT) and intensity modulated radiotherapy (IMRT) have a -technological edge over the conventional 2 dimensional radiotherapy (2DRT), in terms of producing a more conformal radiation dose distribution and a steeper dose gradient leading to better outcomes by sparing the normal tissue adjacent to the targeted area. (8, 9) Although these new health care technologies may hold the promise of providing better health outcomes, they are also a key contributor to rising health care costs. Data from United States, showed that intensity modulated radiation therapy (IMRT) increased the overall costs of breast radiation by 33% from 2001-2005. (10)

In India, expenditure on health care are rising and significant contributor to this expenditure are newer drugs, medical devices and technology. Given the limited public investment in the health sector, it is critical that resources are used wisely on those interventions that are proven to yield best results. Sound investment decisions require technical economic evaluations, which are also one of the ways to guide policy making in terms of rational allocation of resources. A systematic review assessing the extent and quality of evidence of economic evaluations for healthcare in India, found that out of the total economic evaluations (n=104), only 7% were focussed on the cancer care, all of which assessed a cancer screening programme and none had any focus on the treatment/curative component. (11)

The new technologies (IMRT/3CDRT) may be capable of providing better health benefits but may also place an additional financial burden on the health care systems in the provisioning of the same. Despite of the huge spending on the cancer by the Government of India, there is very limited evidence on the cost and no evidence of cost effectiveness in the provisioning of cancer care. Given this background, the present study is designed to assess the cost and cost effectiveness of Intensity Modulated Radiotherapy (IMRT) versus Conventional

Radiotherapy for the treatment of Head and Neck Cancers in India. The present study would provide evidence to the policy makers in terms of essentiality, appropriateness, cost effectiveness and inclusion of newer forms of radiotherapy modalities in the public health system of India.

BACKGROUND

Role of Radiation therapy in HNC treatment

Radiation is a beam of photons, electrons, protons, or neutrons, having the capacity to disrupt the growth of tumour cells by directly or indirectly ionizing the DNA.(12, 13) Radiation therapy is one of the mainstays of treatment of Head and Neck cancers. It is used either with a curative intent, or for palliative treatment, where the main aim is for symptomatic relief. Further, it can be used either alone or in combination with other treatment modalities, i.e., surgery or/and chemotherapy depending upon the state of tumour and clinician judgement. Cancers arising in the head and neck region (oral cavity, tongue, lip, nasopharynx, oropharynx, hypopharynx, larynx, palate, etc.) are in close vicinity to other critical structures such as the brainstem, optic apparatus (eyes, optic nerves, chiasma), parotid glands, lacrimal glands, etc. that makes the radiotherapy treatment challenging and difficult. Although radiation is directed with an intent to maximise the dose to the tumour cell, it is not possible to completely spare the normal tissues surrounding the tumour. External beam radiotherapy is the commonly used form of radiation, which involves, directing a beam of radiation from outside the body towards the tumour inside.(8, 14) The conventional 2 dimensional radiotherapy (2-DRT), delivers radiation in a standard custom shaped through relatively broad beam and provides very limited sparing to the tissue adjacent to the tumour. However, with recent technical advancements and coming up of 3-dimensional conformal radiotherapy (3-DCRT) and Intensity modulated radiotherapy (IMRT) it is possible to specifically target

the tissue mass in relatively higher doses, while minimizing the damage to surrounding tissue.(15-17) While, 3D-CRT uses multiple beams of radiation, which are custom shaped, conforming to the size and dose to the tumour.(15) IMRT in addition, can modulate or control the intensity radiation beam in multiple smaller volumes, to conform more precisely to the three-dimensional shape of the tumour mass. (17)

Status of radiotherapy in India

The National Cancer Control Programme (NCCP) of the country led to the development of 27 regional cancer centres across India for the provision of specialised treatment and undertaking research in India. Further, there are currently 347 radiotherapy centres in the country as per DIRAC (Directory of radiotherapy centres) report. These radiotherapy centres have a total of 347 tele-cobalt and tele-caesium machines (delivers only 2-DRT) and 235 linear accelerators (delivers each of the three modalities).

Evidence on effectiveness

Systematic reviews of studies (both observation and randomised controlled trials) done across the globe have concluded that IMRT, as compared to 2DRT and 3DCRT, by sparing the normal surrounding tissue (parotid glands, pharyngeal constrictor muscles, etc.), while irradiating tumours of HNC, is associated with statistically significant improvements in certain quality of life (QoL) domains, particularly those relating to xerostomia, dry mouth, sticky saliva and eating-related domains.(9, 18, 19) Most of the studies included in these systematic reviews had used cancer specific European Organisation for Research and Treatment of Cancer (EORTC) Quality of Life Questionnaire QLQ-C30; and site specific head and neck questionnaire, i.e., H&N 35 module. (20, 21) There are randomised controlled trials (RCTs) reported from India, comparing IMRT and 3DCRT using cancer specific tools

w.r.t head and neck cancers. (22, 23) But there is no study from India, which had compared IMRT and 2DRT for HNCs for QoL scores. In terms of survival, a meta-analysis comparing all the 3 techniques concluded a non-significant increase in loco-regional control and overall survival in favour of IMRT. Meta-analysis included a study from India, which compared IMRT to 3D-CRT. (9)

Evidence on cost of cancer care

Growing awareness among consumers on advancements in medical technology coupled with rising purchasing power/ disposable income has resulted in inflation of health care expenditures. Unlike in other sectors, the cost of medical technology in the health sector is increasing and contributes to the spiralling healthcare costs.

The literature shows that, there are no studies assessing health system cost of radiotherapy care and only a single study was located, assessing health system cost of surgery. (24) While there are few studies reporting patient level out of pocket expenditures on cancer care. (25-27) One of these studies, done in a tertiary care setting during 2006-07, reported that patients end up paying INR 8184/- for a seven-week course of radiotherapy, 59% of which was spent on transportation, food and lodging. (28) Another study, by analysing NSSO data for the year 2004-05, reported that households affected by cancer spent the equivalent of 36–44% of the annual household expenditures for inpatient expenses and had higher rates of borrowing and selling of assets to fund health-care costs (50%) than control households (16%). (29) Another study, reported that patients with chronic illness such as cancer had a higher risk of incurring catastrophic health expenditure than those diagnosed with a communicable disease.(30)

Evidence on cost effectiveness

There has been no study from India, assessing cost effectiveness of any of the radiation technology either w.r.t head and neck cancer or any other cancer site. But, there are a couple of studies, both from the developed countries, comparing IMRT and 3D-CRT for HNCs. One of these studies, done in the context of Canada, reported that the delivery of IMRT produced 0.48 more QALYs than 3DCRT at an additional cost of \$2447, yielding an incremental cost-effectiveness ratio of \$5084 per QALY gained.(31) Another study from the United Kingdom, reported that in the first 2 years following treatment, IMRT was not cost-effective as compared to 3DCRT, but, over 15 years (considering the lifetime as study horizon), IMRT was more cost-effective at an incremental cost effectiveness ratio of \$34,523 per QALY gained.(32)

Limitations in the existing evidence

In relation to effectiveness, there has been studies from India, which has reported better quality of life with IMRT as compared to 3D-CRT, based on cancer specific QoL questionnaires i.e, EORTC QLQ-C30 and EORTC H&N 35 module. But, for the purpose of an economic evaluation, those health state utility values are needed, which could finally be converted into a single quality of life score. While these disease and site specific questionnaire are more sensitive to measure the specific complications or side effects of treatment, these tools are not capable of measuring the quality of life index score, which is possible with generic QoL health questionnaires such as EQ-5D-DL & SF-36. There are no study from India, which have assessed quality of life index score after HNC treatment with generic tools. In terms of survival, while there has been randomised controlled trials (RCTs) comparing IMRT and 3D-CRT, (22, 23) there are only few observational or single arm studies assessing loco-regional control or overall survival for 2DRT. (33, 34)

In terms of cost, there has been no study from India, measuring the cost of delivering radiotherapy care through different from a health system perspective. Although, some literature is reported on out of pocket expenditures, but these estimates are based on studies done a decade back. Further, none of these studies, measured OOP expenditure specifically related to HNCs. Moreover, there is no study on the cost effectiveness of these treatment modalities from India. Although there are studies on costing and cost effectiveness reported from outside India, but generalizing the results of these studies to Indian settings is a major issue and cannot be done because of the difference in health care costs and health financing system across the countries.

OBJECTIVES

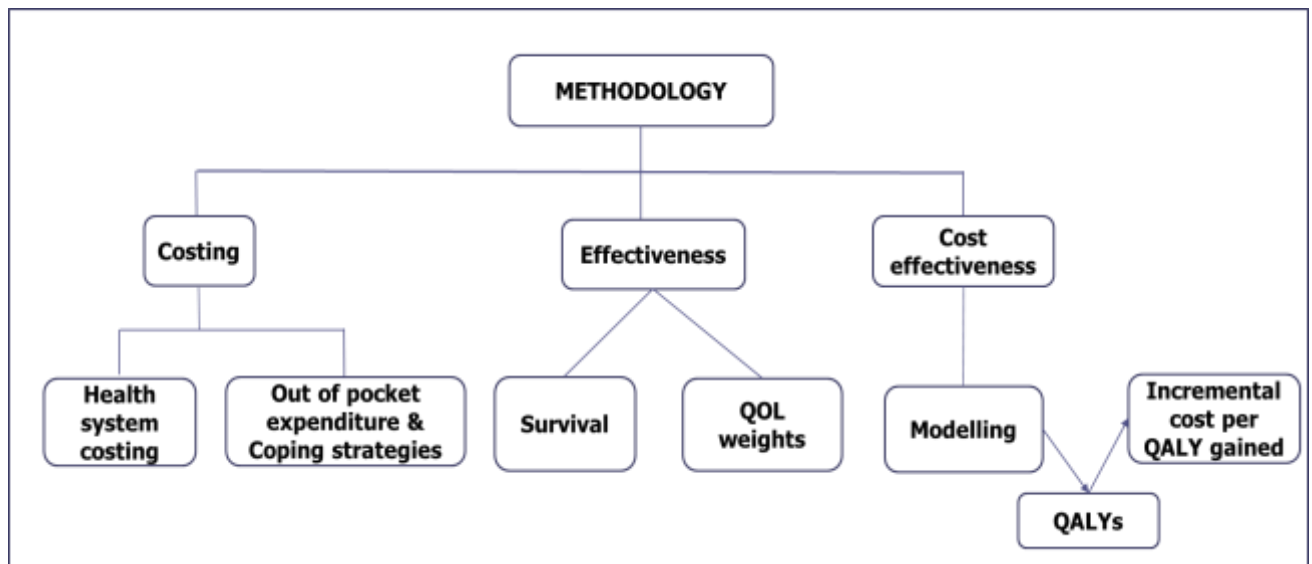
One of the objective of the study will be to estimate the health system cost and out of pocket (OOP) expenditure incurred on Intensity modulated radiotherapy (IMRT) and conventional radiotherapy for HNCs at public sector tertiary care hospital in India. Secondly, the study will assess the effectiveness in terms of survival, incidence of xerostomia and quality of life for HNC patients treated with IMRT and 2DRT. Last and final objective of the study is to estimate the incremental cost per quality adjusted life years (QALY) gained with the use of Intensity modulated radiotherapy (IMRT) as compared to conventional radiotherapy in the treatment of HNC patients.

METHODOLOGY

Framework

Following the objectives of the study, methodological framework has been designed as shown in fig 1. Like any other economic evaluation, the first step requires collecting data on the cost of provision of the intervention and comparator followed by collecting information health outcomes. Finally, there will be development of a Markov model to calculate the quality adjusted life years (QALY) and incremental cost per QALY gained.

Figure 1: Methodological framework



Study area

The present study, a model based cost evaluation, will be conducted in Otolaryngology and Radiotherapy Department of a tertiary care hospital in North India, i.e., Post Graduate Institute of Medical Education and Research (PGIMER), Chandigarh.

Health system costing methodology

Data collection

The economic cost of radiation therapy will be assessed using bottom up costing methods.(35, 36) Cost data will be collected by taking a reference period of financial year 2014-15. The first step involved in the process of cost assessment will be to classify cost centres in terms of direct and indirect cost centres, specifically related to the delivery of radiotherapy care. The next step will be to assess the outputs generated by each cost centres w.r.t the type of service a cost centre delivers. This data on service output (in terms of outpatient consultations, inpatient/bed days, number of patients on IMRT or 2DRT) will be identified by assessing the routine medical records from the outpatient register, inpatient register, radiotherapy procedure register etc.

Following this, inputs required for producing the output in the form of either capital or recurrent resources will be identified and measured. Facility survey will be undertaken to assess the capital resources, i.e. building/space, diagnostic equipment, therapeutic equipment, furniture and other non- medical items. This will be supplemented by reviewing non consumable stock registers. Space area/floor size will be obtained from the engineering wing of the study hospital. Recurrent resources in the form of materials/supplies (masks, stereotactic frame, X-ray films, drugs, injections, office/administrative supplies and other utilities such as water, telephone, electricity, Internet) will be estimated by reviewing the consumables stock register, vouchers, pharmacy records and administrative records. Wage, another recurrent resource, will be estimated firstly by, enumerating or line listing all the staff members, whether medical (consultants, junior and senior residents, nurses, etc.) or non-medical (physicist, medical technicians and engineers, etc.), regular or part time, partly or completely involved in radiotherapy treatment and then by eliciting the salary details of each

staff member from the payslips obtained from the accounts department of the study hospital. After line listing, all the staff members will be interviewed using a semi-structured interview schedule on time allocation for different activities.

Estimates for the rental price, i.e., current market price will be used for assessing the space/building cost. The prices of drugs, consumables and medical equipment will be based on the government contract rate price of the study hospitals. For the furniture and non-medical equipment current market prices will be utilized. A health facility costing questionnaire which has been used in earlier Indian studies, (37, 38) will be adapted in the present setting to be used for data collection in the present study.

Data analysis

Price of the equipment will be annualized (which involves spreading out the costs of capital goods over time periods) over the useful life of the asset to arrive at the equivalent annual cost by taking into consideration the discount rate (time preference for money and inflation) and the lifespan of equipment. (36) For some of the staff members (such as consultants, junior and senior residents, nurses), which are jointly involved in a number of activities (such as OPD/consultation, IPD, planning, dosimetry and delivery of the treatment), cost allocation to the particular activity will be apportioned based on the proportional time contribution towards each of the activities, which will then be multiplied with the gross salary of the staff member. Costs related to water, electricity, laundry, etc. will be apportioned as per floor area. Operating/maintenance and any construction cost will be based on the actual estimated use.

Finally, the total cost of land/space, equipment, drugs, consumables and overheads will be ascertained by multiplying the unit prices of each resource with the quantity consumed,

followed by estimation of unit cost, which will be calculated by dividing the total cost by the service output.

Methodology for assessing out of pocket expenditure and financial risk protection

The present study will follow the Cost of Illness approach for assessing both direct and indirect expenditure incurred by the patients.(39) Patients will be interviewed to assess the direct medical expenditure incurred by them upon on the diagnostic tests, radiotherapy procedures, medicines/drugs, hospitalization if any, user fee, etc. and direct non-medical expenditure in terms of any boarding, loading or food cost. Wage loss of the patient and companion (if any), while coming for radiotherapy treatment will also be assessed in the form of indirect expenditure. Patients will also be interviewed about the various financial strategies undertaken by them to cope up with the out of pocket expenditure in terms of use of current income/savings, borrowing with or without interest, selling of assets or with the use of health insurance of any kind.

Sample size

Sample size calculation for assessing OOP is based on the mean of INR 1062 and standard deviation of 412 as reported from a previous study based in similar settings.(28) By taking INR 40 as level of precision at 95% confidence interval, the sample size come out to be 410.

Data collection tool

A structured interview schedule assessing both the direct and indirect expenditure and coping mechanisms had been developed and has been used in Indian settings.(40) Same tool has been adapted to the present scenario and will be used in the study.

Data collection methods and patient recruitment

The patients will be recruited from the outpatient wing of the Radiotherapy department, and will be interviewed about the expenditure incurred by them on the type treatment i.e., radiotherapy modality they are undergoing. These patients would be recruited based on:

1. Prospective data collection: These patients will be recruited before the start of their radiotherapy treatment and will be interviewed for information on out of pocket expenditure on a daily basis during the period of the radiotherapy course, which lasts for about 5 to 7 weeks. Also, patients will be followed up at 1, 3, 6 and 12 months after the treatment to capture to capture any expenditure incurred after the treatment Receipts, hospital and insurance cost records will be sourced where available from participants to validate reported expenditure. Contact information (address, landline/mobile phone numbers of the patient and at least 2 caregivers/family members) of the study participants will be collected during the baseline interviews. This information will be used for conducting telephonic interviews during the period of follow ups. It has been documented in the literature that telephonic interview works well and has good response rate. (41, 42) Other source of contacting the patient such as through SMS, e-mail along with telephonic interviews woill be adopted to enhance the response rate. Around 1/3rd of the total sampled patients will be recruited prospectively.

2. Retrospective data collection: Out of a total of sample of 410 patients, 2/3rd will be interviewed retrospectively from patients who had completed their radiotherapy within the past one month.

Financial risk protection

The present study will utilise the concept of catastrophic health expenditure (CHE) to measure the extent of financial risk associated with cancer care in public hospitals in India. The WHO concept of CHE will be followed in the study, which defines catastrophic expenditure as the health expenditure >40% of the patient's annual non-food expenditure. To measure the social impact of incurring OOP expenditure in terms of coping strategies undertaken, concept of distress financing shall be applied. Patients using borrowing (both with and without interest) or selling assets as financing options undertaken to cope up with the out of pocket expenditure will be considered suffering from distress financing.

Data analysis

Data analysis will be done using SPSS version 17 (SPSS Inc. Chicago, IL, USA) to assess the OOP expenditure in terms of direct and indirect costs, to identify the risk and intensity of catastrophic expenditure and prevalence of distress financing. Descriptive analysis of OOP expenditure will be done both during radiation therapy and up to 12 months post therapy. Continuous variables will be informed as means $\pm$ standard deviation and categorical variables as proportions (%). Continuous variables will be compared using t tests and one-way ANOVA and categorical variables through chi square test. To estimate indirect costs the human capital approach will be applied, using lost working days and occupation rates. For calculating indirect costs attributable to morbidity, data collected on the income loss and the

working day loss of both the patients and their relatives/carers during the radiotherapy and follow-up period will be used.

Multiple logistic regression analysis a mathematical modelling approach will be performed to examine the differential risk of catastrophic expenditure due to radiation therapy (outcome variable) and prevalence of distress financing (outcome variable) for recruited patients with covariates (independent factors) including socio demographic variables, length of the treatment, presence of insurance, etc. Selection of the covariates will be based on the literature review. The logistic model is useful as the elongated S-shape graph of the logistic function can describe the combined effect of several independent risk factors on the risk of an outcome variable. Measures such as Cox and Snell's R^2 and Nagelkerke's R^2 (or adjusted R^2) will be used to assess the model fit and its extent to which the model predicts all the outcomes perfectly. Sensitivity analysis will be carried out to assess the change in the prevalence of catastrophic expenditure by the varying the cut off from 10% to 40%.

Methodology for effectiveness assessment: Quality of life

Primary data collection will be carried out by interviewing patients for their quality of life and the incidence of toxicities based on the standard tools. Patients with histologically proven head and neck cancer, having a UICC stage of I-Iva, between the age of 18-70 years and on good performance status will be included in the data collection. Further, those patients, who were previously treated for HNC, having uncontrolled co-morbidities, had undergone surgery or unfit for radical treatment will be excluded.

Sample size

Using the results of a previous study by Nutting et al. (43) which reported incidence of grade 2 or worse xerostomia at 12 months as 74% and 38% with the use of 2DRT & IMRT respectively, the sample size (alpha: 0.05, 1-beta: 0.8) was calculated to be 105 considering 36% absolute difference in the incidence of grade 2 or worse xerostomia. Assuming some patients to be deemed in-evaluatable (lost to follow-up or assessments not done), a total of 120 patients will be selected for the study. Thus a total of 24 patients in the IMRT arm and 96 patients in the 2DRT arm will be sampled out for the present study.

Data collection

Patients will be assessed on a weekly basis during the treatment period for collection of data on acute side effects and quality of life. Further, late radiotherapy side-effects will be assessed with the Radiation Therapy Oncology Group (RTOG) scoring system at end of treatment, 6 weeks post radiotherapy followed by at 3 and 6 months post treatment. (44) Similarly, patient-reported quality of life will be assessed through standardised instrument, i.e. with EQ-5D,(45) and also with specific instruments related to the oncology settings, i.e. European Organization for Research and Treatment of Cancer (EORTC) QLQC30 quality-of-life instrument and the associated head and neck specific module HN35 at end of treatment, 6 weeks post radiotherapy followed by every 3 months till 1 year.(20, 21) Patients will also be required to complete these instruments in the clinic before the start of the treatment.

Methodology for effectiveness assessment: Survival rates

A systematic review cum meta-analysis will be undertaken of the existing randomised controlled trials to see the effect of IMRT and 2DRT in terms of incidence of xerostomia, and

Study selection, data extraction and risk of bias assessment will be done by two independent investigators and any disagreement between the two will be sought by the senior investigator. The risk of bias of the included studies will be assessed in accordance with the Cochrane Handbook for Systematic Reviews of Interventions.(46) Based on this, the selected studies would have high, unclear or low risk of bias based on the following items: generation of allocation sequences, allocation concealment, blinding of participants and personnel (performance bias), blinding of outcome assessment (detection bias), incomplete data addressed, presence of biases in reports and other sources of bias that can influence the study's validity.

Review Manager (RevMan) Version 5.3 software will be used for statistical analyses. Pooled risk ratios will be calculated along with 95% CI following random effects model. Further, subgroup analyses will be performed and heterogeneity will be checked using the Chi-square and I square test.

Development of Markov model

A Markov model will be developed based on the natural history of the progression of head and neck cancer and on the review of previously developed models. Development of the model will take into consideration several steps in the form of model parameterization, model calibration, model adaptation and model uncertainty analysis.

A lifetime study horizon will be used to model the costs and consequences in a cohort of 1000 individuals undergoing treatment in the intervention arm i.e., IMRT and the control arm i.e., 2DRT respectively. A societal perspective will be followed, thus, taking into consideration both the health system costs and OOP expenditure. Future costs and

consequences will be discounted at 3%, keeping in view existing methodological guidelines. (36, 47) Further, in sensitivity analysis, different discount rates will be explored.

Extensive review of published literature will be undertaken for obtaining epidemiological and demographic to be used in the model. For the data on health system costs, OOP expenditure incurred in the delivery of treatment with IMRT and 2DRT, estimates based on primary data collection, as explained in the previous section will be used. For information on survival rates, an existing meta-analysis will be updated, the methodology of which is explained above. Further for data on health related quality of life, results from primary data collection will be used.

To assess the impact of uncertainty in the values of parameters used in the model, a probabilistic uncertainty analysis will be done. Under this, various parameters used in the study will be assigned distributions instead of a point value. After assigning distributions, a number of Monte Carlo simulation (say 1000) will be run, allowing for sampling from the various distributions.(48)Thus, for each simulation, different values will be picked from the distributions, and many different ICERs will be calculated. Further, other types uncertainty analysis, i.e., methodological, structural and heterogeneity will be carried out to assess the robustness of incremental cost effectiveness ratio.

Ethical approval

Ethical clearance will be taken from the institute ethics committee of the study hospital. Written informed consent form will be sought from patients for their participation in the data collection for OOP expenditure and quality of life. Also written informed consent will be taken to interview staff for time allocation during the collection of health system cost data.

Outcomes

One of the outcomes of the study will be the estimates on health system cost and OOP expenditure for different treatment modalities related to HNCs, which can further be used for assessment of package rates of these modalities in public sector hospitals of India. Secondly, the study will generate quality of life weights for patients following radiotherapy (IMRT and 2DRT) and pooled survival rates based on the results of meta-analysis. The final outcome of the study will be the incremental cost effectiveness ratio of using IMRT as compared to 2DRT for the treatment of head and neck cancers in India.

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